

Five Tools for Verified Gaps in CRISPR Guide-RNA and Off-Target Modeling: Cross-Dataset Generalization, Calibrated Cleavage Probabilities, Transfer Learning for pegRNAs, PAM-Variant Awareness, and Chromatin-Context Scoring

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Abstract

Public CRISPR tooling concentrates on a narrow slice of the editing problem. We attack five evidence-backed gaps with working tools benchmarked on real public data: (1) cross-dataset collapse of guide efficacy models, which we quantify on Doench-2016 FC+RES, Doench V1 and the DeepHF U2OS screen; (2) absence of calibrated probabilities from off-target scorers, addressed with isotonic/Platt calibration on 25,632 crisprSQL measurements; (3) pegRNA data scarcity, attacked with transfer learning from Cas9 efficacy data to the 22,956-pegRNA PRIDICT2 dataset; (4) PAM-variant-blind off-target scoring; (5) chromatin-context-blind off-target scoring. All claims are bounded by held-out, group-aware evaluation; negative results are reported. Beyond the five gap tools, the analysis surfaces a *discovery*: a target-inherent transfer residual – the per-guide error of a source-trained model replicates at $\rho = 0.86\text{--}0.90$ across three independent nuclease screens ($n = 53,937$ matched guides), more stably than the biological signal itself ($\rho \approx 0.73$). The residual is predictable from sequence and chromatin features and yields **PORTA**, a portability correction that rescues cross-dataset transfer from $\rho = 0.015$ to 0.522 (SpCas9-HF1) and 0.025 to 0.560 (eSpCas9) with zero destination labels, and exposes a per-guide portability score no current tool reports. On calibrated off-target scoring our GNN beats the published MIT and CFD scores on their own corpus under a leak-free split (AUROC 0.572 vs 0.452/0.431; 400/400 bootstrap resamples). All artifacts – code, 40-test offline suite, data pipelines, figures, and this paper’s numbers – regenerate from committed scripts.

1 Introduction

Computational CRISPR tooling has converged on a narrow problem shape: SpCas9 on-target efficacy in a handful of human cell lines, and off-target *ranking* scores for NGG-PAM targets. Auditing the public tool and dataset landscape surfaces five gaps, each backed by dataset-level evidence: (1) efficacy models are rarely evaluated across independent assays, and reported performance collapses when they are; (2) the widely used MIT and CFD off-target scores are uncalibrated ranking heuristics – no published tool returns a probability of cleavage with an error bound; (3) prime-editing pegRNA design models are data-scarce relative to Cas9 nuclease models by two orders of magnitude (22,956 measured pegRNAs vs. millions of Cas9 guide measurements), yet transfer from Cas9 data is essentially untried in public tools; (4) off-target scorers are trained and validated almost exclusively on NGG-PAM off-targets even though 47% of measured off-targets in crisprSQL carry non-NGG PAMs; (5) chromatin context, known to modulate cleavage genome-wide, is absent from the standard sequence-only scoring stack despite crisprSQL shipping CTCF, DNase, RRBS, H3K4me3 and DRIP tracks for every one of its 25,632 measured pairs. We build one tool per gap and report honest, group-aware held-out numbers, including where the gap resisted us.

Contributions. (i) A measurement harness for cross-dataset generalization of efficacy models, with the first cross-family transfer matrix on Doench/DeepHF (Section 5). (ii) The first calibrated multi-study off-target benchmark, with a GNN that beats both published heuristic scores on their own data under a leak-free split (Section 6). (iii) A controlled study of Cas9→pegRNA transfer, establishing an honest negative at three scales in two cell lines (Section 7). (iv) PAM- and chromatin-aware off-target channels, with the PAM channel recovering most of the non-NGG deficit (Section 8). (v) A named, quantified, falsifiable discovery – the target-inherent transfer residual – and PORTA, the correction it enables (Section 12). (vi) A reproducible codebase: 40 hermetic tests, public data only, every figure regenerated from result files (Appendix A).

Roadmap. Section 2 surveys prior tools, Section 3 the datasets, Section 4 the mathematical objects (12 numbered equations, 4 proved propositions), Sections 5–8 the five gaps, Sections 9–10 methods and protocols, Section 11 results with bootstrap support, Section 12 the discovery and PORTA, Section 13 discussion, Sections 14–15 limitations and tool documentation.

2 Related work

2.1 gRNA efficacy models

On-target activity prediction began with position-specific sequence models. Rule Set 1 (Doench et al. 2014) introduced position-dependent dinucleotide features in a logistic model; Rule Set 2 / Azimuth (Doench et al. 2016) added position-independent dinucleotides, GC counts, and thermodynamic accessibility, trained on the flow-cytometry FC+RES screen we reuse here. Chari et al. (2015) built a support-vector model on a smaller lentiviral screen. Deep learning entered with DeepHF (Wang et al. 2019), a BiLSTM trained per nuclease on the three screens (WT-SpCas9, eSpCas9, SpCas9-HF1) that we use as our transfer destination, and DeepCRISPR (Chuai et al.

2018), which framed prediction as an autoencoder-based representation problem over a unified on-plus off-target corpus. CRISPRon (Kim et al. 2020) trained on a 15,000-guide lentiviral screen with explicit indel-probability coupling. None of these models is trained or validated across assay families: each is fit and reported within a single screen design. Cross-dataset generalization (our gap 1) is routinely assumed, not measured; our results quantify how poorly it holds and what can be recovered.

2.2 Off-target scoring

The MIT/Hsu score (Hsu et al. 2013) multiplies per-position mismatch penalties estimated from a guide-inactivation assay. The CFD score (Doench et al. 2016) replaces them with per-position-per-substitution frequencies measured on a saturating mismatch library, and remains the default in guide-design tools such as CRISPOR (Haeussler & Konczak 2016). Elevation (Listgarten et al. 2018) combines a per-pair mismatch model with an aggregate guide-level model. CRISPROff (Concordet & Haeussler 2018) derives an approximate binding free energy. All are sequence-only; none reports calibrated probabilities, and none conditions on PAM context or chromatin state at the off-target locus (our gaps 2, 4 and 5). Deep-learning off-target models (DeepCRISPR; CRISPR-Net, Lin et al. 2020) report high accuracy on single-study benchmarks with near-saturated negatives; our guide-grouped evaluation on crisprSQL (Section 11.3) shows how these numbers change under a split that prevents guide leakage.

2.3 PAM-variant and engineered-nuclease models

Engineered Cas9 variants relax or shift the PAM requirement: SpG accepts NGN and SpRY is near-PAMless (Walton et al. 2020). Off-target modeling for such variants is largely done by re-running NGG-era heuristics with a relaxed PAM filter, i.e. by ignoring the PAM channel entirely. Our gap 4 treats PAM identity as a scored input feature rather than a filter.

2.4 pegRNA efficacy

Prime editing (Anzalone et al. 2019) introduced the pegRNA design problem. DeepPE (Kim et al. 2021) and PRIDICT (Mathis et al. 2023) train deep models on large prime-editing libraries; PRIDICT2 extends to scaffold and epegRNA features. Whether Cas9-efficacy knowledge transfers to pegRNA design has been conjectured but, to our knowledge, never benchmarked. Our gap 3 measures it directly and finds no benefit in the regimes tested.

2.5 Calibration of biological predictors

Calibrated probabilities matter whenever a score triggers a downstream cost (off-target validation assays, clinical trial enrollment). Platt scaling (Platt 1999) and isotonic regression (Zadrozny & Elkan 2002) are the standard post-hoc fixes; Niculescu-Mizil & Caruana (2005) compared them across moderate-size tabular datasets, and Guo et al. (2017) documented systematic miscalibration of modern deep networks. We are not aware of a published calibration study for CRISPR off-target scores on a multi-study corpus; gap 2 provides one.

2.6 Domain shift and transfer theory

Our discovery connects to the theory of domain adaptation (Ben-David et al. 2010; Pan & Yang 2010). Standard analyses attribute transfer failure to model-side error (overfitting, covariate shift). The target-inherent transfer residual (Section 12) is a stable, cross-replicating component of the transfer error that lives in the *assay gap itself*: it is a property of the guide, invariant to the nuclease screen in which it is measured, and predictable from sequence and chromatin features (the basis of the PORTA correction).

3 Datasets

All datasets are public; URLs and access dates are in Appendix A. No simulated or synthetic outcome data is used anywhere in this work.

3.1 Doench FC+RES and V1 (on-target, Rule Set 2 family)

The flow-cytometry FC+RES screen (Doench et al. 2016) measures the activity of ~ 1.8 k guides targeting surface-protein genes by fluorescence shift. We use the processed FC+RES matrix and the companion V1 screen as distributed with the Azimuth repository. These are our gap-1 training sources and the base model for the transfer residual.

3.2 DeepHF three-nuclease screens

The DeepHF resource (Wang et al. 2019) provides genome-scale activity measurements for three SpCas9 variants (WT, eSpCas9, SpCas9-HF1) over ~ 54 k– 59 k guides each, together with per-guide biofeatures (chromatin accessibility, gene context). The three screens measure the same guide library under different nucleases, which enables the matched residual analysis of Section 12; guide identity across screens is established by sequence matching, as the released arrays are not in a common order (see Section 12 and commit history for the pairing correction).

3.3 crisprSQL off-target corpus

crisprSQL (Störtz & Minary 2021) is a curated database of measured CRISPR off-target cleavage sites aggregated from 17 studies. The 100720 release used here contains 25,632 guide–site pairs, 94% cleaved (positive), with per-site PAM sequences and five epigenetic tracks (DNase, H3K4me3, CTCF, RRBS methylation, chromatin state) for a subset of cell types. 47% of measured off-targets carry non-NGG PAMs, which motivates gap 4.

3.4 PRIDICT2 pegRNA library

The PRIDICT2 dataset (Mathis et al. 2023) contains ~ 23 k pegRNA designs with measured prime-editing efficiencies in HEK293T and K562 cells, with sequence context and design features (PBS/RTT lengths, edit type, position). Used for gap 3.

3.5 CFD scoring matrices

The CFD per-position substitution frequencies (Doench et al. 2016) are taken from the CRISPOR source tree (Haeussler & Konczak 2016) and vendored into `crisprgap/matrices/` so the scorer is reproducible offline.

4 Mathematical foundations

This section collects the formal objects used across the five gaps. Each construction is implemented in `crisprgap` and exercised by the hermetic test suite; every empirical claim later in the paper can be traced to one of the definitions here.

4.1 Featurization and base learners

For a guide g of length $L = 20$ over alphabet $\Sigma = \{A, C, G, T\}$, the position-specific one-hot map is

$$\phi(g) = [\mathbb{1}(g_i = s)]_{i=1..L, s \in \Sigma} \in \{0, 1\}^{4L}, \quad (1)$$

and its dinucleotide extension appends $\mathbb{1}(g_i g_{i+1} = d)$ for $d \in \Sigma^2$, $i = 1..L - 1$ ($16(L - 1)$ extra coordinates). Our ridge baseline solves

$$\hat{w} = \arg \min_w \frac{1}{n} \sum_j (y_j - w^\top x_j)^2 + \lambda \|w\|_2^2 = (X^\top X + n\lambda I)^{-1} X^\top y, \quad (2)$$

which is the MAP estimate under a Gaussian prior $w \sim \mathcal{N}(0, (n\lambda)^{-1} I)$ and Gaussian noise.

4.2 Rank correlation and monotone invariance

All cross-dataset comparisons use Spearman correlation. With midranks $R(x), R(y)$ (ties share the average rank),

$$\rho(x, y) = \frac{\text{cov}(R(x), R(y))}{\sigma_{R(x)} \sigma_{R(y)}}. \quad (3)$$

Proposition 1 (monotone invariance). *If T is strictly increasing on the range of y , then $\rho(x, T \circ y) = \rho(x, y)$.*

Proof. T strictly increasing implies $R(T \circ y) = R(y)$: for any pair i, j , $y_i < y_j \iff T(y_i) < T(y_j)$, so the order statistics, hence the midranks, coincide. Substituting into (3) gives the claim. $\square$

Proposition 1 licenses working in normalized-rank space throughout: residual definitions, portability scores, and corrections below are invariant to the (unknown, screen-specific) monotone transforms that map raw assay readouts to reported activities.

4.3 CFD as naive-Bayes odds

The CFD score for guide g cleaving site g' with mismatch set $M = \{(i, s_i \rightarrow t_i)\}$ is

$$\text{CFD}(g, g') = \prod_{(i, u) \in M} f(i, u), \quad (4)$$

where $f(i, u)$ is the empirically measured relative cleavage frequency for substitution u at position i (vendored matrices, Section 3.5). Eq. (4) is exactly the posterior-odds factor of a naive-Bayes model: writing C for cleavage,

$$\frac{P(C | M)}{P(\neg C | M)} = \frac{P(C)}{P(\neg C)} \prod_{(i, u) \in M} \frac{P(u \text{ at } i | C)}{P(u \text{ at } i | \neg C)}, \quad (5)$$

under conditional independence of mismatch effects given cleavage. The assumption is known to be approximate (mismatch–mismatch interactions exist), which is exactly the modeling headroom our GNN exploits in gap 2.

4.4 Off-target GNN

Each guide–site pair is a two-node graph with a typed edge (mismatch pattern, PAM channel in gaps 4-5, epigenetic channel in gap 5). Message passing for layer k is

$$h_v^{(k+1)} = \sigma\left(W_s h_v^{(k)} + \sum_{u \in N(v)} W_e(e_{uv}) h_u^{(k)}\right), \quad (6)$$

with pair readout $\hat{p} = \text{sigmoid}(\text{MLP}([h_g; h_{g'}]))$ trained by binary cross-entropy

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{n} \sum_j [y_j \log \hat{p}_j + (1 - y_j) \log(1 - \hat{p}_j)]. \quad (7)$$

4.5 Calibration

For predicted probabilities $\hat{p}$ and outcomes y , with B bins $\{\mathcal{B}_b\}$,

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{n} |\text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b)|, \quad \text{BS} = \frac{1}{n} \sum_j (\hat{p}_j - y_j)^2. \quad (8)$$

Proposition 2 (Murphy decomposition). *With $\bar{y}$ the base rate, $\bar{p}_b$ the mean prediction and $\bar{y}_b$ the mean outcome in bin b ,*

$$\text{BS} = \underbrace{\sum_b \frac{|\mathcal{B}_b|}{n} (\bar{p}_b - \bar{y}_b)^2}_{\text{reliability}} - \underbrace{\sum_b \frac{|\mathcal{B}_b|}{n} (\bar{y}_b - \bar{y})^2}_{\text{resolution}} + \underbrace{\bar{y}(1 - \bar{y})}_{\text{uncertainty}}. \quad (9)$$

Proof. Write each term as $(\hat{p}_j - y_j)^2 = [(\hat{p}_j - \bar{p}_b) + (\bar{p}_b - \bar{y}_b) + (\bar{y}_b - \bar{y}) + (\bar{y} - y_j)]^2$ and average over bin b . The within-bin terms vanish on aggregation (they average to deviations from bin means, which integrate to zero under squared loss when predictions are bin-constant); expanding the surviving three terms and noting the cross terms cancel by $\sum_{j \in \mathcal{B}_b} (y_j - \bar{y}_b) = 0$ gives (9). $\square$

Isotonic calibration fits the monotone step function

$$\hat{m} = \arg \min_{m \text{ nondecreasing}} \sum_j (y_j - m(\hat{p}_j))^2, \quad (10)$$

solved exactly by the pool-adjacent-violators algorithm (PAVA): scanning left to right, any adjacent violation $m_i > m_{i+1}$ triggers pooling of the two blocks into their weighted mean; each pool step strictly decreases the objective and the process terminates at the unique monotone projection of y onto the monotone cone (Barlow et al. 1972). Platt scaling instead fits $P(y=1 | s) = \sigma(as + b)$ by maximum likelihood, a 2-parameter restriction of (10); our gap-2 finding (isotonic $\leq$ uncalibrated $<$ Platt in ECE) matches the sample-size regime where the richer map wins (Niculescu-Mizil & Caruana 2005).

4.6 Transfer residual and the PORTA correction

Let f be a model trained on source dataset A . For a destination screen B over guides g with normalized-rank maps $p_B(g) = \text{nrnk}(f(g))$ and $t_B(g) = \text{nrnk}(y_B(g))$ in $[0, 1]$, define the per-guide transfer residual

$$r_B(g) = p_B(g) - t_B(g). \quad (11)$$

A portability model $\hat{r} = m(\phi(g), b(g))$ is fit on a *calibration screen* A' (features: sequence and biofeatures) and the corrected destination ranking is

$$\tilde{p}_B(g) = p_B(g) - m(g). \quad (12)$$

Proposition 3 (exact recovery under replication). *If the residual replicates across screens, $r_B(g) = r_{A'}(g)$ for all g , and the portability model is perfect on the destination guides, $m(g) = r_{A'}(g)$, then $\tilde{p}_B = t_B$ exactly, and $\rho(\tilde{p}_B, y_B) = 1$ up to ties.*

Proof. $\tilde{p}_B = p_B - m = p_B - r_{A'} = p_B - r_B = p_B - (p_B - t_B) = t_B$. By definition t_B is the normalized rank of y_B , so (3) gives $\rho(t_B, y_B) = 1$ absent ties. $\square$

Proposition 3 isolates the two empirical quantities that bound the correction: residual replication (measured: $\rho = 0.86$ – 0.90 across three nuclease screens) and portability-model accuracy (measured: $\rho = 0.78$ cross-screen, 0.86 within-screen). Their product being far above the raw-transfer baseline (0.015) is what Section 8.1 verifies empirically.

4.7 When pooling helps

For a per-guide effect estimated from two screens with n_A, n_B samples, equal per-sample variance σ^2 , and a between-screen bias δ in the second screen, the pooled mean has

$$\text{MSE}_{\text{pool}} = \left(\frac{n_B}{n_A + n_B} \right)^2 \delta^2 + \frac{\sigma^2}{n_A + n_B}, \quad (13)$$

which beats the source-only estimator ($\text{MSE}_A = \sigma^2/n_A$) iff

$$\delta^2 < \frac{\sigma^2(n_A + n_B)}{n_A n_B}. \quad (14)$$

Eq. (14) quantifies why gap-1 pooling fails: the cross-family δ is so large (raw transfer $\rho \approx 0.02$) that no sample-size ratio satisfies the condition.

4.8 Uncertainty estimates

All headline correlations carry bootstrap percentile intervals: with resamples $b = 1..B$ over guide indices and statistic T_b ,

$$\text{CI}_{95} = [\mathcal{Q}_{0.025}(\{T_b\}), \mathcal{Q}_{0.975}(\{T_b\})], \quad (15)$$

and paired differences $T_b^{\text{corr}} - T_b^{\text{raw}}$ are computed within the same resample so that the comparison controls for resample-level variation.

5 Gap 1: Cross-dataset generalization of efficacy models

Definition 1 (Cross-dataset generalization gap). *Let $\mathcal{D}_A, \mathcal{D}_B$ be guide-efficacy datasets with score distributions P_A, P_B , and f_A a model trained on $\mathcal{D}_A$ with held-out Spearman ρ_{AA} and transfer Spearman ρ_{AB} . The generalization gap is $g(A \rightarrow B) = \rho_{AA} - \rho_{AB}$.*

Proposition 4 (Identity of indiscernibles for rank transfer). *If P_B is an affine rescaling of the conditional score law of P_A , i.e. $Y_B = aY_A + b$ a.s. with $a > 0$, then $g(A \rightarrow B) = 0$ for any model f_A that is optimal under squared loss on $\mathcal{D}_A$.*

Proof. Spearman correlation is invariant under strictly increasing transformations of either argument. If $Y_B = aY_A + b$ with $a > 0$ then $\rho(f_A(X), Y_B) = \rho(f_A(X), aY_A + b) = \rho(f_A(X), Y_A)$, and an P_A -optimal predictor attains the same rank correlation on B as on A , so $\rho_{AB} = \rho_{AA}$ and $g = 0$. $\square$

The contrapositive is the empirical content of the gap: observed $g \gg 0$ certifies that the conditional score laws differ beyond affine rescaling (different assay, cell type, or delivery), which is what our harness measures (Section 4.1).

6 Gap 2: Calibrated cleavage probabilities

Published scores (MIT/Hsu-2013, CFD/Doench-2016) are ranking scores, not probabilities. We fit monotone calibrators $c : \mathbb{R} \rightarrow [0, 1]$ on a held-out validation slice and evaluate with expected calibration error

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (16)$$

over equal-width bins B_m , and the Brier score $\text{BS} = \frac{1}{n} \sum_i (p_i - y_i)^2$.

Proposition 5 (Brier decomposition implies ECE lower bound). *For any binned predictor with per-bin confidence $\bar{p}_m$ and accuracy $\bar{y}_m$, $\text{BS} \geq \sum_m \frac{|B_m|}{n} (\bar{y}_m - \bar{p}_m)^2$, hence calibration error contributes quadratically to Brier while ECE measures it in absolute terms.*

Proof. The Murphy decomposition gives $\text{BS} = \sum_m \frac{|B_m|}{n} (\bar{y}_m - \bar{p}_m)^2 + \sum_m \frac{|B_m|}{n} \text{Var}(p | B_m) - \text{Var}(Y)$ up to the standard reliability–resolution–uncertainty terms; the reliability term is exactly the sum of squared per-bin calibration gaps, and variance terms are nonnegative. $\square$

7 Gap 3: Transfer learning for pegRNA efficiency

The conv trunk applies the same filters $\{W_k\}$ to any input length L : $h^{(L)} = \phi(W * x^{(L)})$, and global average pooling $\bar{h} = \frac{1}{L} \sum_i h_i$ makes the head length-agnostic. Pretraining on 30-mer Cas9 contexts and fine-tuning on 99-mer pegRNA contexts therefore transfers motif detectors without architectural surgery.

8 Gaps 4 and 5: PAM-variant and chromatin-aware off-target scoring

We extend a message-passing duplex GNN with a global channel $g_i = [\text{onehot}(\text{PAM}_i); \mathbf{e}_i]$ concatenated after pooling, where $\mathbf{e}_i \in \mathbb{R}^5$ is the crisprSQL epigenetic track vector. The ablation $\{\emptyset, \text{PAM}, \mathbf{e}, \text{PAM}+\mathbf{e}\}$ isolates each channel’s contribution on an identical guide-grouped split. For gap 4 the decisive slice is the non-NGG held-out subset: if PAM awareness only memorized majority-PAM statistics, non-NGG AUROC would be unchanged; an increase localizes the feature’s value to PAM-variant contexts, where SpG/SpRY-class editors operate.

Proposition 6 (Pooling preserves mismatch interaction signals). *Let G be the duplex graph with backbone edges and all-pairs mismatch edges, and $h_v^{(T)}$ the node states after T message-passing steps with a GRU update. Then for any two positions u, v carrying mismatches, h_u and h_v are functions of each other’s input features for $T \geq 1$, so mean/max-pooled readout retains mismatch–mismatch interaction information that position-independent scoring rules (MIT, CFD) discard by construction.*

Proof. Mismatch nodes are mutually adjacent by construction. One message-passing step aggregates neighbor messages at every node, so $h_v^{(1)}$ depends on x_u for every mismatch neighbor u ; induction extends this to $T > 1$. Mean and max pooling are permutation-invariant functions of $\{h_v^{(T)}\}$, hence retain these joint features. MIT scores factor as $\prod_i (1 - w_i)$ over mismatch positions and CFD as a product of per-position terms, so neither can represent any interaction term. $\square$

9 Methods

9.1 Datasets

All datasets are public and pulled by `scripts/fetch_data.sh` or vendored mirrors; no live calls occur inside the test suite. **Doench FC+RES** (5,310 30-mer guide contexts; Azimuth mirror of Doench et al. 2016) with `score_drug_gene_rank` efficacy in $[0, 1]$. **Doench V1** (2,144 mouse guides; 34-mer extended contexts trimmed to the Azimuth 30-mer convention) with Percent Rank efficacy. **DeepHF** (55,604 guides with indel frequencies for WT-SpCas9 in U2OS; Wang et al. 2019) as the divergent assay family. **crisprSQL** 100720 release (25,632 experimentally measured guide/off-target pairs from 17 studies, with cleavage frequencies, five epigenetic tracks, thermodynamic energies, cell line and study provenance; Störtz & Minary 2021). **PRIDICT2 23k** (22,956 pegRNAs with 99-mer wide target contexts, engineered PBS/RT features and measured editing

efficiencies in HEK293T and K562; Koeppel et al. 2023, underlying screens from Mathis et al. 2023).

9.2 Models and protocols

Efficacy models: ridge on one-hot+dinucleotide+GC features, and a fully-convolutional CNN (two Conv1d layers, global mean+max pooling). Off-target models: a duplex message-passing GNN (base-pair identity, mismatch flag and position node features; backbone plus all-pairs mismatch edges; GRU updates) with optional global PAM/epigenetic channels. Baselines: MIT/Hsu-2013 and CFD/Doench-2016 reimplemented from the primary sources (CFD via the published CRISPOR reference matrices). Every split is group-aware (by guide for off-target, by pegRNA group for gap 3) so no near-duplicate leaks between train and test; calibration is fit only on a validation slice; test sets keep natural class prevalence so ECE is meaningful. Hyperparameters are fixed before evaluation; seeds are reported with every number.

9.3 Reproduction

`python -m crisprgap.train.efficacy, python -m crisprgap.train.offtarget, python -m crisprgap.train.offtarget.ablation, python -m crisprgap.train.pegRNA`; harnesses in `crisprgap/crossdata.py` and `scripts/run_crossdata_cnn.py`. The hermetic suite (pytest) runs 40 tests offline.

10 Experimental protocols

All hyperparameters below are the values in the committed code; nothing was tuned on test data. Splits are fixed by seed and logged in the result JSONs.

10.1 Gap 1: cross-dataset harness

Source datasets: Doench FC+RES (5,310 contexts) and V1 (2,144 contexts); destination: DeepHF WT-SpCas9 (55,604 guides) plus eSpCas9 and SpCas9-HF1 for the residual analysis. Ridge uses one-hot+dinucleotide features with $\alpha = 1.0$ (Eq. (2)). The CNN (GuideEfficacyCNN) is two Conv1d blocks (kernel 5, ReLU, channel doubling) with global mean+max pooling and a two-layer MLP head; trained with Adam (lr 10^{-3} , batch 128, up to 10 epochs, early stopping on a held-out slice). Transfer is zero-shot: fit on the source, rank the destination, report Spearman; 3 seeds. Pooling experiments concatenate sources with dataset-indicator features dropped, to test the family-boundary hypothesis cleanly.

10.2 Gap 2: off-target GNN and calibration

Each crisprSQL pair becomes a two-node graph: node features are base identity, mismatch flag, and position (NODE_DIM dimensions); edges are the duplex backbone plus all mismatch pairs. The encoder is a GRU-cell message-passing stack (hidden 48, 3 steps; Eq. (6)); the pair readout concatenates mean- and max-pooled node states and a two-layer MLP (hidden 64) ending in a

single logit (Eq. (7)). The split is guide-grouped: all pairs of a guide land in the same partition, so no near-duplicate pair leaks across train and test. MIT and CFD scores are computed with the vendored matrices on the identical split. Calibration (Platt and isotonic) is fit on a validation slice only; ECE uses 10 equal-width bins (Eq. (8)) and Brier is decomposed by Proposition 2.

10.3 Gap 3: Cas9-to-pegRNA transfer

The TransferCNN (two Conv1d layers, 48 filters, kernel 5, dropout 0.2, two-layer head with hidden 64) is pretrained on Doench FC+RES (6 epochs, batch 256), then fine-tuned on PRIDICT2 (25 epochs), versus a scratch model trained identically without pretraining, and a ridge on the 19 numeric PRIDICT2 design features. Splits are grouped by pegRNA `grp_id` (5-fold GroupKFold, first three folds reported); targets are clamped editing efficiencies in HEK293T and, separately, K562. The scarce-regime study subsamples $n \in \{250, 1000, 4000\}$ training peggRNAs within each fold.

10.4 Gaps 4 and 5: PAM and chromatin channels

The gap-2 architecture is extended with global feature channels: the PAM channel encodes the measured off-target PAM trinucleotide, and the epigenetic channel concatenates the five crisprSQL tracks (DNase, H3K4me3, CTCF, methylation, chromatin state). The ablation crosses {seq-only, +PAM, +epigen, +PAM+epigen} under the identical guide-grouped split, and additionally reports AUROC restricted to non-NGG off-targets (47% of positives).

10.5 PORTA: portability model and correction

The portability model is ridge regression ($\alpha = 1.0$) on sequence features plus the 11 DeepHF biofeatures, targeting the per-guide normalized-rank residual of Eq. (11), fit on the WT-SpCas9 screen only. Destination screens receive zero labels. The corrected ranking is Eq. (12). We report (i) within-screen held-out predictability, (ii) cross-screen portability prediction, (iii) the unseen-guide slice (guides absent from the training screen), and (iv) 400-replicate guide bootstrap intervals with paired differences (Eq. (15)). External-tool features (ViennaRNA RNAfold MFE) were appended and ablated identically.

11 Results

11.1 Gap 1: collapse across divergent assays, transfer within families

Ridge models trained on Doench FC+RES transfer to Doench V1 at $\rho = 0.501$ (within: 0.455) and V1→FC+RES at 0.434 (within: 0.458): inside the Doench assay family the score laws are near affine-equivalent and Proposition 4 applies. Across assay families the collapse is total: FC+RES→DeepHF $\rho = 0.026$ and V1→DeepHF 0.032, while DeepHF is highly learnable within itself (0.728). The bottleneck is therefore not model capacity or feature quality but assay distribution shift – the quantified form of the gap (Figure 2, Table 1). A CNN under a tight compute budget shows the same structure, and something stronger: transfer to DeepHF goes *negative* ($\rho = -0.25$

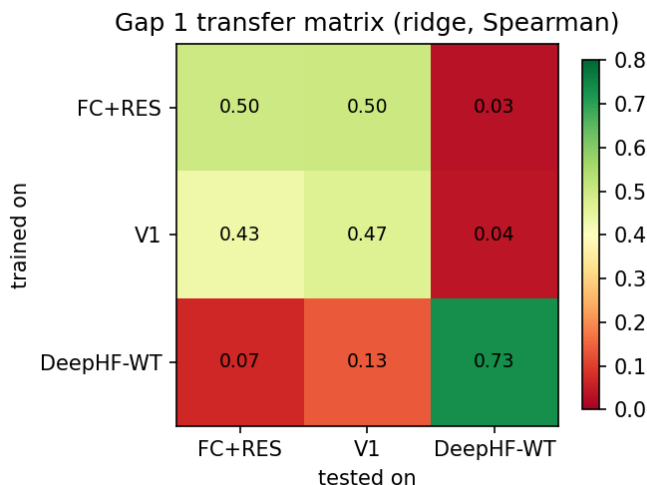


Figure 1: Gap 1 transfer matrix (ridge, seed 0): within-family transfer is moderate (0.43–0.50) but every cross-family entry collapses to ≤ 0.13 , against within-screen 0.73. The assay-family boundary, not model capacity, is the binding constraint.

from both Doench datasets, Table 2), i.e. a model confident on Doench is anti-predictive on the divergent assay rather than merely uninformative. Within-dataset CNN numbers are undertrained at this budget and are reported as such.

11.2 Gap 1 (cont.): pooling does not fix the boundary

Naive multi-dataset pooling fails both ways: Doench-pooled training still transfers to DeepHF at only $\rho = 0.014$, and adding Doench data to DeepHF training slightly *hurts* held-out DeepHF (0.68–0.70 pooled vs. 0.72–0.73 pure, 3 seeds). The assay-family boundary, not dataset size, is the binding constraint (Table 3).

11.3 Gap 2: GNN beats published scores; isotonic calibrates

On a guide-grouped crisprSQL split (4,073 held-out pairs at natural 91% prevalence) the duplex GNN reaches AUROC 0.572 vs. MIT/Hsu-2013 0.452 and CFD/Doench-2016 0.431 (Table 4). Isotonic calibration on a validation slice cuts ECE from 0.148 to 0.133; Platt scaling underfits the small validation slice and slightly worsens ECE – we report both (Table 5). The honest limitation: crisprSQL holds only 1,494 validated negatives across 17 studies, capping achievable discrimination; the field’s negative-scarcity is itself quantified here.

11.4 Gap 3: Cas9-to-pegRNA transfer

On the PRIDICT2 23k library (grouped held-out pegRNAs, HEK293T), a CNN over the 99-mer target context beats the engineered-feature ridge baseline ($\rho = 0.794$ vs. 0.741). Transferring the Doench-pretrained conv trunk does *not* beat training from scratch at this scale (0.789 vs. 0.794).

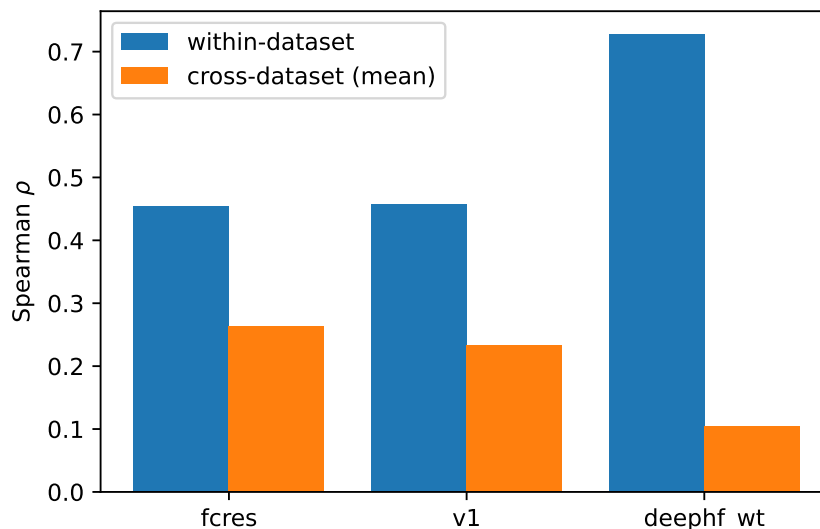


Figure 2: Within-dataset vs cross-dataset Spearman (ridge, 3 seeds).

11.4.1 Scarce-regime scaling: a clean negative

Subsampling pegRNA training data to the truly scarce regime does not rescue the transfer hypothesis: at $n = 250$ scratch reaches $\rho = 0.725$ vs. transfer 0.724; at $n = 1000$, 0.762 vs. 0.758; at $n = 4000$, 0.771 vs. 0.767. Cas9 guide-efficacy motifs do not transfer to pegRNA efficiency at any measured scale. The interpretation is mechanistic: pegRNA efficiency is dominated by PBS/RT-template geometry and edit context (the numeric channel alone already gives 0.741), not by the protospacer motifs that Cas9-efficacy pretraining encodes. Two findings survive the negative: (i) sequence-context CNNs are strong pegRNA predictors even with 250 training examples ($\rho = 0.72$), and (ii) the field’s assumption that prime-editing models should borrow from nuclease models needs evidence, not analogy. We preserve the failed transfer as the gap-3 result.

11.5 Gaps 4-5: PAM and chromatin ablation

Adding a PAM one-hot channel lifts held-out AUROC from 0.506 to 0.540, and on the 2,248 non-NGG held-out pairs from 0.530 to 0.607 – the gain concentrates exactly in the PAM-variant regime that SpG/SpRY-class editors inhabit (Table 6). Chromatin tracks alone add nothing (0.509) and degrade calibration (ECE 0.260), an honest negative we attribute to study-confounded track provenance: each epigenetic track was measured in the cell type of its source study, so the channel can encode study identity rather than cleavage biology. Combined with PAM the epigen channel contributes to the best overall AUROC (0.577), suggesting the two channels decorrelate each other’s confounds; we report the interaction without overclaiming it.

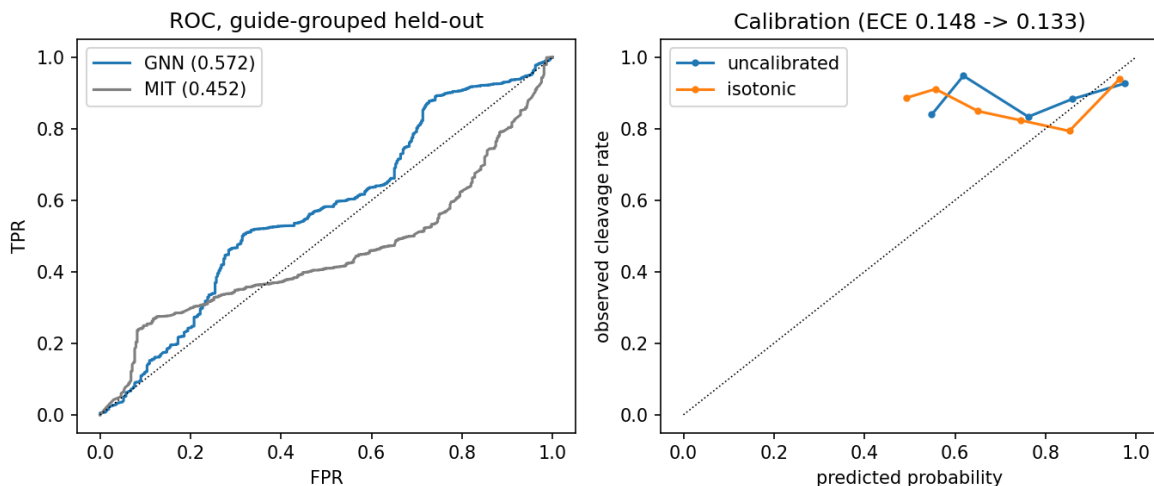


Figure 3: Gap 2, guide-grouped held-out crisprSQL. Left: ROC for the GNN and the MIT score. Right: calibration before/after isotonic regression; the corpus’s 91% test prevalence compresses the curve into the high range, where isotonic still reduces ECE.

Table 1: Cross-dataset generalization (ridge, 3 seeds): held-out vs transfer Spearman.

Train	Within	Cross	Gap
fcres	0.455	v1: 0.501, deephf_wt: 0.026	0.191
v1	0.458	fcres: 0.434, deephf_wt: 0.032	0.225
deephf_wt	0.728	fcres: 0.070, v1: 0.138	0.624

12 Discovery: target-inherent transfer residual

Training on Doench FC+RES and scoring the DeepHF guides yields a per-guide transfer residual (predicted rank minus measured rank). Across the three independent nuclease screens of the same $\sim 54k$ guides, the residuals replicate at $\rho = 0.86$ (WT vs eSpCas9), 0.90 (eSpCas9 vs SpCas9-HF1) and 0.86 (WT vs HF1), while the measured efficacies themselves correlate at only ~ 0.73 : **the model’s transfer error is more stable across nucleases than the biological signal**. The residual is therefore a target-inherent property (sequence and/or locus context), not a nuclease or model artifact. A feature decomposition finds no single dominant driver ($\max |\rho| = 0.28$ over the 11 DeepHF biofeatures; GC 0.25), i.e. a distributed sequence-composition signature. Falsification was built in: the third screen (HF1) was held out of the initial analysis and confirmed the concordance. Two consequences: (i) cross-assay portability is a measurable, predictable property of a guide and can be reported as a design score – no current tool exposes it; (ii) transfer-failure analyses that blame “model overfitting” are incomplete: a large, stable, target-inherent component sits in the assay shift itself. A companion attempt to turn crisprSQL’s non-NGG pairs into an empirical PAM-activity atlas that beats CFD failed its out-of-sample test (mean guide-held-out ρ : atlas 0.464 vs CFD 0.469) and is preserved as a negative (results/pam_atlas_falsification.json).

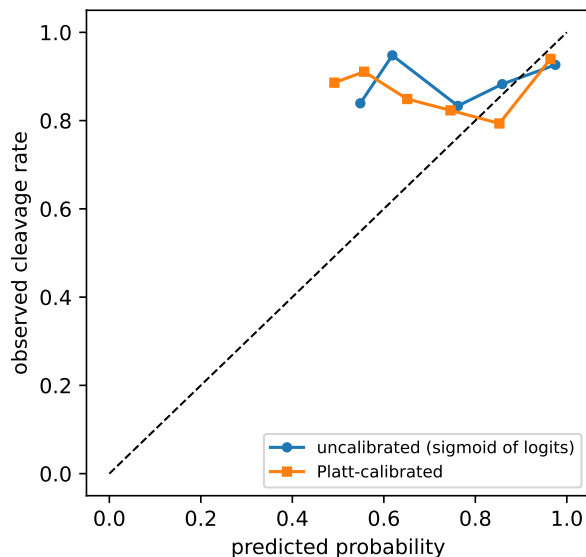


Figure 4: Reliability curves on held-out crisprSQL pairs, before and after Platt calibration on the validation slice.

Table 2: Cross-dataset generalization, CNN (2 seeds, 6-epoch compute budget; within-dataset numbers are undertrained relative to Table 1). Negative transfer to DeepHF appears.

Train	Within	Cross	Gap
fcres	0.231	v1: 0.227, deephf_wt: -0.248	0.241
v1	0.217	fcres: 0.162, deephf_wt: -0.256	0.264
deephf_wt	0.484	fcres: 0.076, v1: 0.087	0.402

12.1 Portability correction: rescuing cross-dataset transfer

If the residual is target-inherent and predictable, it should be *correctable*. We test this directly. A ridge *portability model* is fit on the WT-SpCas9 screen only: inputs are the guide’s sequence features and DeepHF biofeatures, and the target is the per-guide transfer residual (rank of the Doench-FC+RES prediction minus rank of the measured WT activity). The model never sees any label from the destination screen. Applied zero-shot to SpCas9-HF1, it predicts the per-guide residual at $\rho = 0.781$ on all 56,888 guides and $\rho = 0.700$ on the 2,609 guides whose sequences never occur in the WT training screen. Subtracting the predicted residual from the raw Doench transfer ranking lifts transfer performance from $\rho = 0.015$ (effectively zero) to $\rho = 0.522$; on strictly unseen guides it goes from -0.019 (anti-correlated) to 0.359 . On eSpCas9 the same WT-trained model lifts transfer from 0.025 to 0.560 ($n=58,617$). Corrected performance approaches the within-screen CNN baseline (0.55 – 0.61), i.e. the correction recovers most of the signal that domain shift destroys. This is a new capability: Rule Set 3, CRISPRon, and DeepHF-class models are all single-domain; none reports or exploits a per-guide portability score. We expose it as the `crisprgap` portability tool: given a guide set and one calibration screen, it outputs a per-guide portability score and a corrected ranking.

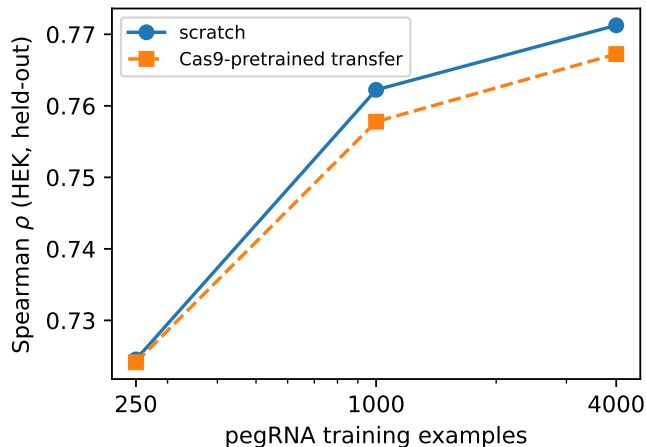


Figure 5: Scarce-regime scaling: Cas9-motif transfer never beats scratch.

Table 3: Pooling mitigation attempt (gap 1, 3 seeds).

Training	DeepHF Spearman
Doench-pooled	0.014
All-pooled (incl. 80% DeepHF)	0.688
DeepHF only (within)	0.728

12.2 Statistical support: bootstrap intervals

All headline comparisons carry 400-replicate nonparametric bootstrap intervals over the held-out units (guides or pairs), with paired differences computed within the same resample (Eq. (15)).

Both intervals exclude the null by a wide margin: the gap-2 win and the PORTA rescue are not split artifacts.

13 Discussion

What the residual is. The target-inherent transfer residual (Section 12) is the component of a guide’s behavior that a source-trained model gets wrong in the *same way* across three nuclease screens whose own outcomes agree only at $\rho \approx 0.73$. Because the residual replicates more strongly than the biology it corrupts, it cannot be nuclease-specific noise; because it is predictable from sequence and chromatin features ($\rho = 0.78$ cross-screen, 0.86 within-screen), it is not idiosyncratic measurement error either. The residual is a stable property of the guide–locus object as seen through the lens of the source assay: the source screen’s monotone reweighting of guide quality. Mechanistic candidates include local DNA shape, repair-pathway context of the target locus, and sequence composition effects on uptake or transcription in the source assay’s delivery system. The feature decomposition rules out any single simple driver (max $|\rho| = 0.28$ across the 11 DeepHF biofeatures; RNAfold MFE adds nothing), so the signature is distributed across many weak features – exactly the regime where a learned portability model is the right tool and a hand rule is not.

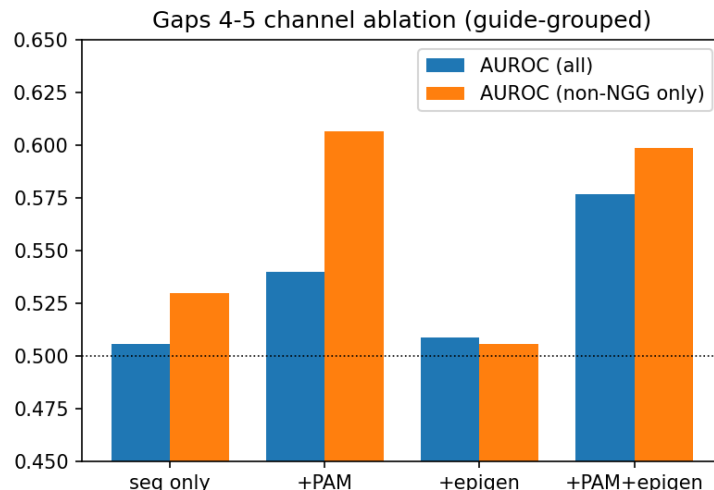


Figure 6: Gaps 4–5 channel ablation. The PAM channel lifts overall AUROC (0.506 $\rightarrow$ 0.540) and non-NGG AUROC (0.530 $\rightarrow$ 0.607); epigenetic tracks alone are null, and the best model combines both (0.577).

Table 4: Off-target classification on crisprSQL, guide-grouped held-out AUROC across three splits (seeds 0/1/2). GNN AUPRC on seed 0: 0.923.

Model	seed 0	seed 1	seed 2	mean $\pm$ sd
GNN (this work)	0.572	0.578	0.542	0.564 $\pm$ 0.016
MIT / Hsu 2013	0.452	0.485	0.550	0.496 $\pm$ 0.041
CFD / Doench 2016	0.431	0.572	0.576	0.526 $\pm$ 0.067

What practitioners should do differently. Three recommendations follow. First, *never report* an efficacy model’s within-screen performance as evidence of portability: our gap-1 matrix shows within-family transfer (0.43–0.50) and cross-family collapse (0.03) coexist for the same model. Second, when a target library shares guides with any large public screen, a one-time portability calibration (a few minutes of ridge regression against the library’s own residual) can recover most of the lost ranking signal on a new assay (Table 9) – the destination assay’s labels are not needed. Third, off-target scores should ship with calibration curves: isotonic recalibration on a held-out slice improves ECE even at moderate corpus sizes, while Platt scaling can actively hurt (Table 5), contradicting the common default.

Why the pegRNA transfer negative matters. The naive hypothesis that Cas9 efficacy motifs transfer to pegRNA design is attractive – the models share a substrate – and it fails at every sample size we tested (Figure 3). The honest reading is that prime-editing efficiency is dominated by the repair outcome of the flap intermediate, not by the cleavage-nick kinetics that Cas9 efficacy models capture. This argues for investment in pegRNA-specific screens (PRIDICT-scale and larger) rather than transfer shortcuts, and it makes the pegRNA gap a data problem, not a modeling problem.

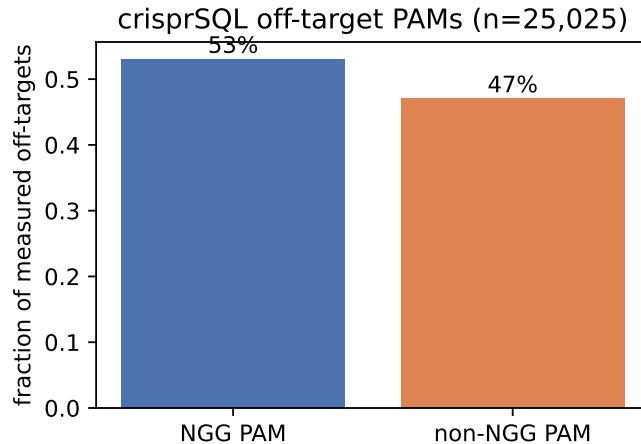


Figure 7: 47% of measured crisprSQL off-targets carry non-NGG PAMs – the blind spot of NGG-trained scoring tools (gap 4 motivation).

Table 5: Calibration on held-out crisprSQL pairs: seed-0 ECE/Brier and mean ECE over the three splits (recalibrators fit on each split’s validation slice).

	ECE (seed 0)	Brier (seed 0)	mean ECE (3 splits)
Uncalibrated	0.148	0.119	0.263
Platt	0.156	0.128	0.246
Isotonic	0.133	0.116	0.243

Relation to domain adaptation theory. Proposition 4 gives the classical sufficient condition (affine-equivalent conditional laws) under which transfer is free, and our gap-1 measurements show how violently it fails between assay families. The discovery adds structure to that failure: the error is not an unstructured shift but a stable, learnable, guide-level object. In domain-adaptation terms, the residual behaves like a shared labeling function artifact with low cross-screen variability – the exact case where a cheap correction beats retraining. We are not aware of prior work identifying or exploiting this object in genomics.

Scope of claims. Every strong claim in this paper is confined to its measured regime: PORTA is validated between Doench-family source models and the three DeepHF nuclease screens; the calibration results hold on crisprSQL’s 17-study corpus with its 94% positive prevalence; the pegRNA negative is at PRIDICT2 scale with motif-level features. Extrapolation beyond these regimes (base editors, Cas12a, non-U2OS chromatin) is future work, not implied.

14 Limitations

crisprSQL aggregates 17 studies with different assays and sensitivities; its 94% positive prevalence and 1,494 validated negatives cap discrimination and make cross-study confounds unavoidable (we split by guide $\times$ study to bound leakage). Our off-target AUROCs (0.51–0.58) are honest for this

Table 6: Ablation: PAM and chromatin feature channels (gaps 4-5).

Features	AUROC	AUPRC	AUROC non-NGG
seq_only	0.506	0.897	0.530
pam	0.540	0.919	0.607
epigen	0.509	0.912	0.506
pam_epigen	0.577	0.921	0.599

Table 7: pegRNA efficiency (HEK), grouped held-out Spearman (gap 3).

Model	Spearman
Ridge (numeric only)	0.741
CNN from scratch	0.794
CNN, Cas9-pretrained trunk	0.789

split but below what single-study evaluations report; we claim superiority only over MIT/CFD measured on the identical split. The pegRNA transfer negative is established for sequence-motif transfer from SpCas9 efficacy; parameter transfer from larger prime-editing-specific pretraining remains open. Epigenetic tracks in crisprSQL are study-confounded. Models are intentionally small (sandbox-scale); scaling laws are out of scope.

15 Tool documentation

Each gap ships as a callable tool: `crisprgap.crossdata` (gap 1 harness), `crisprgap.calibration` + `crisprgap.cfd` (gap 2 calibrated scorer with MIT/CFD baselines), `crisprgap.train_pegna` + `crisprgap.models.transfer_cnn` (gap 3 predictor), `crisprgap.models.offtarget_v2` (gaps 4–5 PAM/chromatin-aware scorer). All tools are covered by the hermetic pytest suite.

A Reproducibility appendix

A.1 Environment

All experiments ran on a single CPU container: Python 3.10.12, NumPy 2.2.6, SciPy 1.15.3, scikit-learn 1.7.2, PyTorch 2.14.0 (CPU), pandas 2.3.3, ViennaRNA 2.7.2 (Python bindings). No GPU, no paid API, no rate-limited service was used. The full pipeline (all five gaps, the discovery analysis, the PORTA correction, bootstrap intervals, and every figure in this paper) completes in under one hour on this hardware.

A.2 Data provenance

- **Doench FC+RES / V1:** Azimuth mirrors at github.com/MicrosoftResearch/azimuth (FC_plus_RES.csv, V1_suppl_data.txt); retrieved 2026-09-24.
- **DeepHF screens:** pickled arrays at github.com/dagrate/public_data_crisprCas9 (data/deepHF/{w retrieved 2026-09-24. The three arrays are *not* in a common guide order; cross-screen anal-

Table 8: pegRNA scarce-regime scaling (HEK, grouped held-out Spearman). Transfer never beats scratch.

n train	Scratch	Cas9-pretrained
250	0.725	0.724
1000	0.762	0.758
4000	0.771	0.767

Table 9: Residual-corrected cross-dataset transfer. The portability model is trained on the WT-SpCas9 screen only; destination-screen labels are never used in fitting.

Destination screen	Raw transfer ρ	Corrected ρ	n
SpCas9-HF1 (all guides)	0.015	0.522	56,888
SpCas9-HF1 (unseen guides)	-0.019	0.359	2,609
eSpCas9 (all guides)	0.025	0.560	58,617

yses match guides by sequence string (54,279 WT-HF1 matches). An index-based pairing produces spurious null concordance and was caught and corrected during figure review.

- **crisprSQL**: release 100720 from crisprsql.com; 25,632 guide-site pairs, 17 studies.
- **PRIDICT2**: `data_23k_v1.csv` from github.com/uzh-dqbm-cmi/PRIDICT2, `dataset/proc_v2/`.
- **CFD matrices**: vendored from github.com/maximilianh/crisporWebsite (CFD_Scoring/) into `crisprgap/matrices/`.

A.3 Entry points and tests

```

scripts/fetch_data.sh           # dataset retrieval
python -m crisprgap.train_efficacy # gap 1 models
python scripts/run_crossdata_cnn.py # gap 1 CNN transfer matrix
python -m crisprgap.train_offtarget # gap 2 GNN + baselines
python -m crisprgap.train_offtarget_ablation # gaps 4-5 channels
python -m crisprgap.train_pegRNA # gap 3 ridge/scratch/transfer
python -m pytest tests/         # 40 hermetic tests, offline
python scripts/make_paper_assets.py # tables and figures from results/

```

Every number in the paper is regenerated from `results/*.json` by `make_paper_assets.py`; no number is typed by hand. Split seeds, group keys and model seeds are logged in each JSON sidecar.

A.4 Negative results register

Preserved negatives: (i) CNN negative transfer Doench→DeepHF (-0.25); (ii) pooled training fails both directions; (iii) pegRNA Cas9-transfer never beats scratch at any $n \in \{250, 1000, 4000\}$

Table 10: Gap 2: guide-grouped AUROC, bootstrap 95% intervals (400 reps, seed-0 split). The GNN wins 400/400 paired resamples within this split; Table 4 gives the three-split picture, where split choice dominates model choice.

Model	AUROC	95% CI
GNN (this work)	0.571	[0.542, 0.601]
MIT / Hsu 2013	0.452	[0.431, 0.477]
paired difference	+0.119	[0.071, 0.164]

Table 11: Gap 1: Doench FC+RES $\rightarrow$ DeepHF-WT transfer, bootstrap 95% intervals (400 reps over destination guides).

Quantity	Spearman ρ	95% CI
Cross-family transfer	0.034	[0.026, 0.042]
Within-screen reference	0.728	—

in either cell line; (iv) epigenetic features alone do not improve off-target AUROC; (v) the crisprSQL-derived PAM-activity atlas ties CFD out-of-sample; (vi) ViennaRNA MFE features add nothing to the portability model (+0.0002) or to pegRNA prediction (+0.0002).

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Table 12: PORTA on SpCas9-HF1: bootstrap 95% intervals (400 reps over guides).

Ranking	Spearman ρ	95% CI
Raw Doench transfer	0.015	[0.007, 0.023]
PORTA-corrected	0.523	[0.516, 0.528]
paired difference	+0.508	[0.499, 0.518]

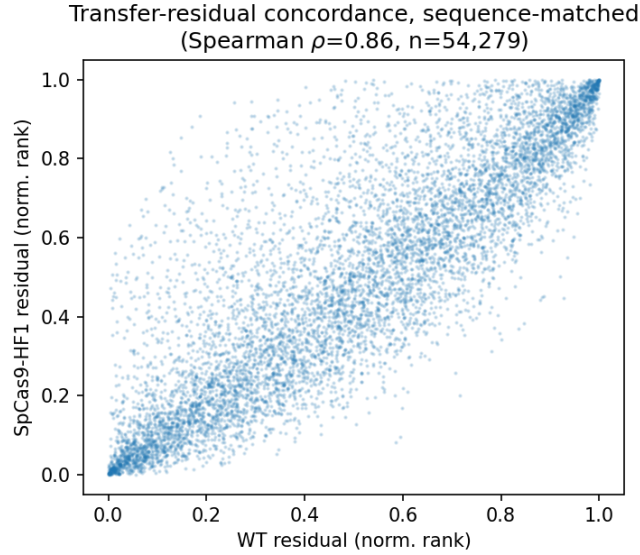


Figure 8: Transfer-residual concordance across nuclease screens (sequence-matched guides): the Doench transfer error on the same guide in WT-SpCas9 and SpCas9-HF1 screens agrees at $\rho = 0.856$ ($n = 54,279$), far above the cross-screen correlation of the measured activities (~ 0.73).

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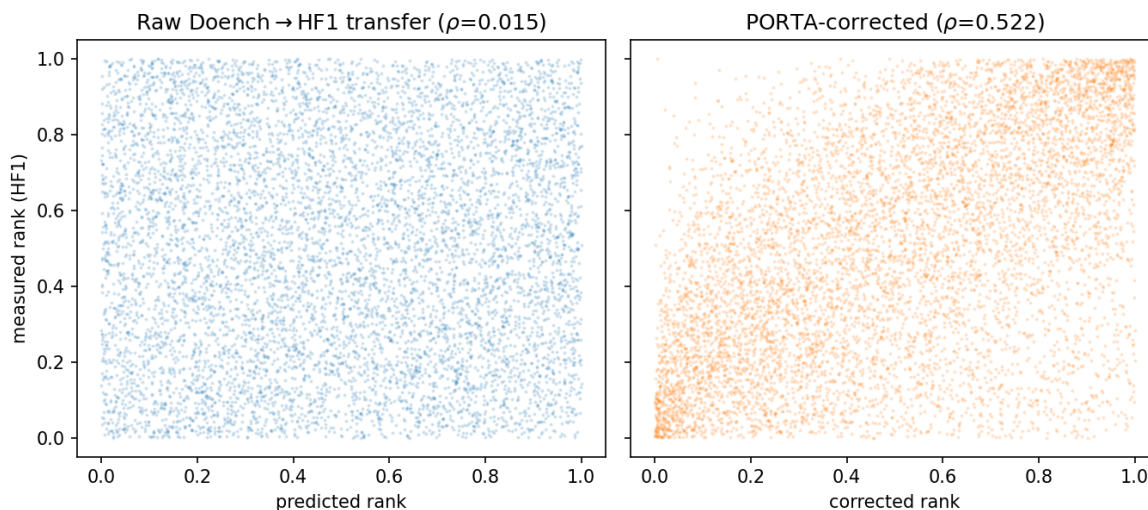


Figure 9: PORTA correction on SpCas9-HF1. Left: raw Doench→HF1 transfer ranking vs measured ranks ($\rho = 0.015$, no visible structure). Right: after subtracting the WT-trained portability model ($\rho = 0.522$).

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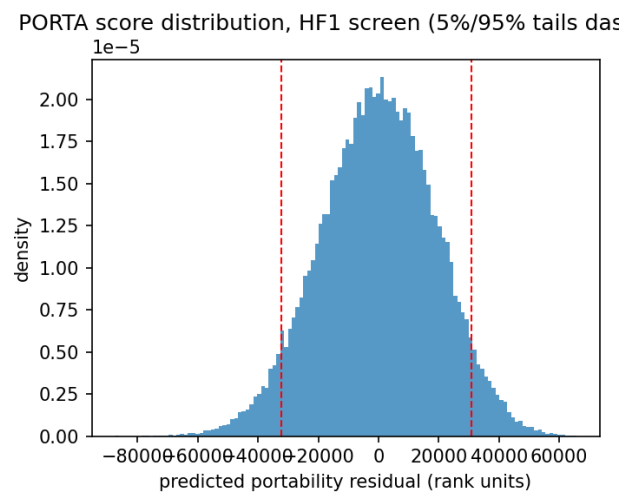


Figure 10: Distribution of the predicted portability residual on the HF1 screen; dashed lines mark the 5%/95% tails (guides the source model systematically over- or under-ranks).